Elements, Compounds and Mixtures

Elements are pure substances made up of only one kind of atom. They cannot be broken down into simpler substances. Compounds are pure substances made up of one kind of molecule. They can be broken down into simpler substances. Mixtures are not pure substances and can be separated into their various components.

1. Write symbols for these elements (see how many you can do without using your periodic table).

Element	**Symbol** (have a go without the periodic table)	**Symbol** (check the periodic table)	**Element**	**Symbol** (have a go without the periodic table)	**Symbol** (check the periodic table)
Aluminium			Carbon		
Copper			Hydrogen		
Lead			Manganese		
Nickel			Oxygen		
Silver			Sulfur		
Calcium			Chlorine		
Gold			Iron		
Magnesium			Mercury		
Nitrogen			Silicon		
Sodium			Zinc		

2. Indicate if the substances below are elements, compounds or mixtures:

Iron _____
Air _____
Salt solution _____
Oxygen _____
Carbon dioxide _____
Water _____
Iron and sulfur _____
Sulfur _____

3. Chemists make us of prefixes in naming compounds. What do these prefixes mean?

mono _____ di _____ tri _____ tetra _____ penta _____

4. Write chemical formulas for these compounds:

Carbon monoxide _____
Sulfur trioxide _____
Phosphorous pentachloride _____
Sulfur dioxide _____
Carbon tetrachloride _____
Dinitrogen tetraoxide _____

EXTENSION QUESTION

5. Complete the table below:

Formula	Name of compound	Number and type of atom
CO_2		
NO_2		
NaCl		
CS_2		
HCl		
$CuSO_4$		
$CaCO_3$		